

Exploiting LES computations and machine learning to improve wind turbine wake reproduction in URANS

Andrea Carloni^a, Giovanni Delibra^a, Valerio Francesco Barnabei^a, Alessandro Corsini^a, Lorenzo Tieghi^b, and Filippo De Girolamo^c

^aSapienza University of Rome, Rome, Italy

^bTrento University, Trento, Italy

^cKTH Royal Institute of Technology, Stockholm, Sweden

Abstract

URANS computations are widely reported as unreliable in reproducing the development of wind turbine wakes, and thus have limited applications in farms, where wake interactions are of uttermost importance in simulating the behavior of downstream devices.

In this manuscript we explore the possibility of using a pipeline of k-means clustering and neural network regression to derive a machine learnt correction of eddy viscosity in URANS. This correction is based on data from high-fidelity LES computations and used to derive a local regression of eddy viscosity, specific to different portions of the wind turbine wake that is automatically segmented at runtime.

LES, RANS and “machine learnt-enhanced RANS” computations are carried out in OpenFOAM-v22.12, modelling the NREL 5 MW reference turbines with the Actuator Line Method and accounting for a synthetic atmospheric boundary layer at the inflow.

Then a clustering methodology is applied to automatically segment the wake into different regions at runtime as in [1, 2].

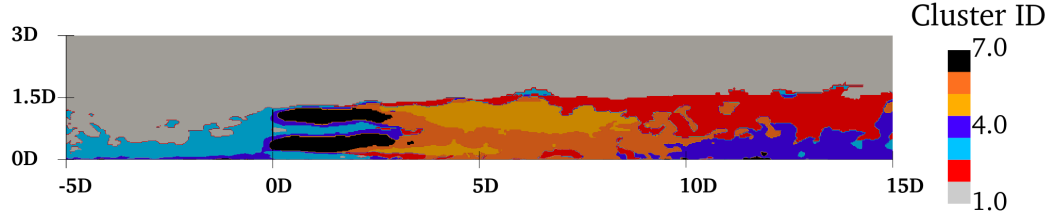


Figure 1: Example of the wake decomposition via clustering [1].

A data-driven correction of the RANS eddy viscosity is introduced, for every cell, as an additive term,

$$\nu_{t,ML} = \nu_{t,RANS} + \Delta\nu_{t,ML,i}, \quad i = 1, \dots, N_c, \quad (1)$$

Where N_c is the total number of clusters. The ML term $\Delta\nu_{t,ML,i}$ is therefore conditioned on the cluster label, allowing different eddy-viscosity corrections to be applied in different wake regions. The trained models are embedded in a modified PIMPLE solver, where the eddy viscosity is updated iteratively within the turbulence-correction loop.

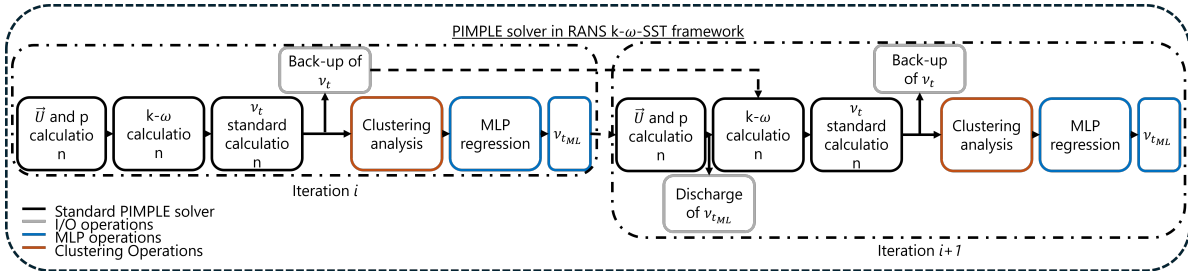


Figure 2: Pipeline of the modified PIMPLE solver with ML-based correction.

The results show that the ML-based RANS model improves wake-flow prediction over the baseline RANS closure in both near and far wake regions. Tests on different mesh resolutions indicate that accurate predictions can be retained on coarser grids,

up to 7 times fewer cells, leading to substantial computational savings with respect to LES. Clustering and non-dimensional topological features improve physical interpretability, supporting model trustworthiness and potential generalization to different flow conditions and turbine configurations.

References

- [1] Andrea Carloni, Giovanni Delibra, Filippo De Girolamo, Valerio Francesco Barnabei, and Lorenzo Tieghi. Data-driven decomposition of horizontal-axis wind-turbine wakes. In *ASME Turbo Expo 2026*, Milan, Italy, 2026.
- [2] Lorenzo Tieghi, Davide Cerbarano, Filippo De Girolamo, Valerio Barnabei, and Giovanni Delibra. A novel methodology for the automatic decomposition of hawt wakes with k-means clustering. *Wind Energy*, 28, 05 2025.